

The effects of free-stream turbulence and angle of attack on the aerodynamics of a cylinder with rectangular 5:1 cross section

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ARTICLE INFO

Keywords:

BARC
Rectangular cylinder
Wind tunnel tests
Turbulent flow
Angle of attack
Impinging shear-layer instability

ABSTRACT

The aerodynamics of a stationary two-dimensional cylinder with rectangular 5:1 cross section is the object of an international experimental and numerical benchmark study (BARC). The unsteady flow past such a bluff body is governed by the impinging shear-layer instability mechanism, which is also responsible for potential flow-induced vibrations. This paper discusses the characteristics of the high-Reynolds-number unsteady flow over the cylinder, investigated in the wind tunnel through pressure and force measurements on a sectional model in smooth and turbulent flows. In addition, hot-wire anemometry measurements of the velocity in the wake of the cylinder were carried out in smooth flow. The study is particularly focused on understanding the influence of free-stream turbulence and angle of attack on the aerodynamic behavior of the body. Key features are highlighted, such as a slight Reynolds-number dependence of force coefficients and Strouhal number in smooth flow, or the turbulence-induced variation of the length of the separation bubbles along the lateral sides of the cylinder. Incoming turbulence is also found to affect the vortex-shedding mechanism of impinging shear-layer instability, promoting the switch from a dominant mode to another. Finally, the wake measurements described some vortex-shedding features, confirming a slight Reynolds-number dependence of the Strouhal frequency.

1. Introduction

The rectangular shape is a recurring geometry in civil, military and industrial structures, and therefore it has often been the object of studies concerning its aerodynamics and proclivity to flow-induced vibrations. For instance, tall buildings are often shaped as elongated prisms and two-dimensional rectangular cylinders have commonly been considered as qualitative schematizations of bridge decks (Matsumoto, 1996). Many complex-shaped structures also present slender elements with rectangular cross sections. In addition, two-dimensional rectangular cylinders have often been considered as a reference to model the complex aerodynamics of rectangular under-slung helicopter loads (Greenwell, 2011; Prosser and Smith, 2016).

For a theoretical cross section with perfectly sharp corners and smooth surfaces, the side ratio alone defines the rectangular geometry, which, due to its simplicity, clearly highlights interesting fluid dynamic phenomena and allows a detailed study of the aerodynamic features of the generated flow field. Moreover, depending on the side ratio and turbulent characteristics of the incoming flow, rectangular prisms are prone to different types of flow-induced excitation, such as vortex-induced vibration (e.g. Komatsu and Kobayashi, 1980; Shiraishi and Matsumoto, 1983; Nakamura and Nakashima, 1986; Naudascher and

Wang, 1993; Marra et al., 2011, 2015), hard- and soft-type transverse galloping (e.g. Parkinson, 1965; Novak, 1972; Nakamura and Tomonari, 1977; Washizu et al., 1978), interference of vortex-induced vibration and galloping (e.g. Parkinson and Wawzonek, 1981; Bearman et al., 1987; Mannini et al., 2014, 2016), torsional galloping (or torsional flutter; e.g. Nakamura and Mizota, 1975; Matsumoto et al., 1997) and coupled flutter (e.g. Matsumoto, 1996; Bartoli and Righi, 2006). Consequently, it is extremely important from the practical engineering point of view to investigate the characteristics of flow past these simple geometries and to understand those mechanisms that can produce vibrations of elastic structures.

For these reasons, the BARC benchmark was launched in 2008 during the 6th International Colloquium on Bluff Body Aerodynamics and Applications (Bartoli et al., 2008), focusing on the stationary rectangular 5:1 cylinder, where the short side of the section faces the flow. This side ratio is particularly interesting, since in smooth flow the shear layers separating at the leading edges are known to reattach intermittently close to the trailing edges (as for all rectangular cylinders with side ratio between 3.2 and 7.6 in smooth flow, according to Parker and Welsh (1983)), so that a strong sensitivity of its aerodynamics on several geometrical features and flow conditions is expected. In addition, Ohya et al. (1992) noted that the vortex-shedding mechanism

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is less regular (and then challenging to predict) than for slightly smaller (such as 3:1 and 4:1) or larger (such as 6:1) side ratios.

A synopsis of the experimental and numerical results obtained in the first four years of benchmark activity has been recently published (Bruno et al., 2014). Therein, the good agreement between results in terms of near-wake flow, base pressure and drag coefficient was shown. By contrast, the comparison highlighted a significant dispersion of flow features and pressure distribution along the lateral sides of the cylinder. A collection of previous experimental results for drag coefficient and Strouhal number can be found in Mannini et al. (2011).

Among the main results achieved during the benchmark, it is worth reminding the measurements in a high-pressure wind tunnel performed by Schewe (2013), who investigated the Reynolds-number range between 4000 and 600,000 (based on the cross-flow section dimension). He found only a slight change of the main global aerodynamic parameters for a null flow incidence but a significant variation of the mean lift coefficient with the Reynolds number for small angles of attack (an increase of about 100% and 70% in the tested Reynolds-number range respectively for incidences of 2° and 4°). In particular, a quick change of mean lift coefficient and Strouhal number is clear for Reynolds numbers below about 10,000. In this respect, it is to note that for a rectangular 3:1 cylinder Okajima (1982) showed a strong Reynolds-number dependence of the Strouhal number for values below about 5000. The Reynolds-number effects observed by Schewe were explained by the two-dimensional unsteady Reynolds-averaged Navier-Stokes (URANS) computations carried out by Mannini et al. (2010), who emphasized the key role played by the separation bubble on the lateral side of the section where shear-layer reattachment occurs, the length of which reduces while increasing the Reynolds number. In that work, the Authors also highlighted the fact that the vortex shedding in the case of an angle of attack of 4° was much more regular at a Reynolds number of 370,000 than 5200.

Other interesting numerical studies were conducted in the framework of the BARC benchmark, discussing also technical aspects related to the numerical approach. In particular, Bruno et al. (2010) carried out a three-dimensional large-eddy simulation (LES), which enabled the identification of five topological mean flow structures around the cylinder (“inner region”, “recirculation region”, “main vortex”, “reattached flow” and “reversed flow in the wake”). In addition, the mechanisms of vortex formation close to the upstream sharp edges, vortex coalescence and convection are described in details. Mannini et al. (2011) performed three-dimensional detached-eddy simulations (DES) and emphasized the crucial influence on the results of the amount of numerical dissipation introduced to stabilize the central-difference discretization scheme and of the spanwise extension of the computational domain. The latter issue was further investigated through LES computations in Bruno et al. (2012) together with the effect of refining the mesh in the spanwise direction. The results pointed out the great importance of these two parameters for the accuracy of the results. Recently, Patruno et al. (2016) concentrated on the influence of small angles of attack on the resulting flow field and aerodynamic quantities, comparing URANS and LES numerical strategies. Ricci et al. (2016) studied the sensitivity of LES results to a low level of incoming turbulence. Finally, Mariotti et al. (2016) carried out URANS simulations with aleatory parameters (angle of incidence, and free-stream longitudinal turbulence intensity and length scale) and investigated the probabilistic propagation of the uncertainty.

For the BARC section, and in general for rectangular cylinders with side ratios in the range between 3 and 15, the vortex-shedding mechanism is not of Kármán type, i.e. triggered directly by the interaction between upper and lower shear layers in the region behind the body, but it is due to the “impinging shear-layer instability” or “edge tone” (Rockwell and Naudascher, 1979; Nakamura et al., 1991; Ohya et al., 1992). This phenomenon is also known as “impinging leading-edge vortices” (Naudascher and Wang, 1993) and is produced by the “impinging” of a shear layer on an obstacle downstream, namely

the afterbody in the trailing edge region in the case of rectangular cylinders. This generates a feedback control through pressure pulses that travel upstream at the speed of sound. This feedback may be an order of magnitude stronger than that in the absence of impingement and is able to trigger large flow fluctuations at the leading edge. In fact, Nakamura and Nakashima (1986) showed that both vortex shedding and vortex excitation are possible for rectangular, H and T shapes also in the presence of a splitter plate, thus meaning that a single shear layer can be unstable in the presence of a downstream corner. Nevertheless, in the absence of a splitter plate, the separation bubble grows and envelops the trailing edge, allowing the fluid within the bubble to pass into the vortex formation region behind the body (Parker and Welsh, 1983). Clearly, the interference with the vortex forming on the opposite side of the cylinder has an effect on the vortex street.

Tan et al. (1998) and Hourigan et al. (2001), through two-dimensional and three-dimensional direct numerical simulations, and Mills et al. (2002, 2003), through flow visualizations and PIV measurements in wind and water tunnels, studied the interaction phenomena between leading- and trailing-edge vortices and their phasing in rectangular cylinders with long afterbody. Nakamura et al. (1996) experimentally observed transition from Kármán vortex street to impinging shear-layer instability mechanism for a Reynolds number around 250 (350 according to two-dimensional direct numerical simulations). In addition, impinging shear-layer instability exhibits different modes of vortex shedding depending on the side ratio, which in some cases can also coexist (Nakamura et al., 1991; Ohya et al., 1992; Naudascher and Rockwell, 1994). As a consequence, the Strouhal number presents evident jumps in correspondence of the side ratios for which a switch to a higher vortex-shedding mode occurs.

Another key issue that deserves great attention is the role played by free-stream turbulence in the aerodynamics of the BARC section. In these respects, one of the earlier investigation on the effects of incoming turbulence on the flow around bluff bodies was carried out by Vickery (1966), while Mulhearn (1973) visualized the reattachment lines on a square cylinder and highlighted their different location in smooth and turbulent flow. Kiya and Sasaki (1983) undertook a thorough study of the steady and unsteady characteristics of the separation bubble in the case of a long plate with blunt front corners. In their review, Bearman and Morel (1983) identified three basic mechanisms by which free-stream turbulence and the high-Reynolds-number mean flow over bluff bodies interact: i) anticipated transition to turbulence in the shear layers; ii) enhanced mixing and entrainment, which produce an increased growth of the free shear layers and a more pronounced curvature inward towards the body, leading to a stronger tendency to reattach; iii) distortion of free-stream turbulence by the mean flow. In the particular case of rectangular cylinders with sharp edges, the first mechanism was recognized to play a minor role. The thicker shear layers in turbulent flow were visualized by Laneville et al. (1977) for a rectangular 2:1 cylinder. Gartshore (1973), Nakamura and Ohya (1984) and Nakamura and Ozono (1987) distinguished between the effect of small-scale turbulence, i.e. comparable with the thickness of the shear layers, and large-scale turbulence, i.e. comparable with the body size. The former enhances the mixing and the entrainment of fluid from the near wake and accelerates the growth of the shear layers promoting their earlier reattachment; in contrast, the latter weakens regular vortex shedding by reducing the spanwise correlation of the process. This distinction is in line with the analysis of Gerrard (1966), who noted that there are two simultaneous characteristic lengths that contribute to the definition of vortex-shedding frequency for a bluff body, namely the scale of the formation region and the thickness of the shear layers when they diffuse. Haan Jr. et al. (1998) emphasized the importance of length scales of the incoming turbulence of the order of one tenth of the cross-section depth.

In the literature, a great effort has also been devoted to study the effect of the integral length scale of free-stream turbulence (Hillier and Cherry, 1981; Courchesne and Laneville, 1982; Bearman and Morel,

1983; Nakamura and Ozono, 1987; Li and Melbourne, 1995; Haan Jr. et al., 1998; among others). Although it does not seem possible to draw final conclusions, one may dare to claim that, for values significantly larger than the body depth, the mean aerodynamic properties of the bluff body flow are fairly insensitive to this parameter, whereas the fluctuating quantities are much more dependent on it.

The overview given by Bearman and Morel (1983) underscores the lacuna in the literature concerning the effects of free-stream turbulence on flows over rectangular cylinders with intermittent reattachment of the separated shear layers. From this point of view, the studies of Courchesne and Laneville (1982), Li et al. (1995) and Haan Jr. et al. (1998) respectively on the 3:1, 4:1 and 6.67:1 sections merit special attention. Nevertheless, in the Writers' opinion the dynamics of vortex shedding and impinging shear-layer instability in turbulent flow is a subject that still requires further investigation, as it also emerges from the discussion in Nakamura and Ozono (1987).

The present paper reports the results of wind tunnel measurements of unsteady surface pressures, forces and wake flow velocities on a stationary sectional model. Given the expected high sensitivity of the aerodynamic behavior of the rectangular 5:1 cylinder to several experimental conditions, particular care was devoted to realize a large model of very high quality (prismatic shape, sharp edges, smooth surfaces, absence of screws). This ensures a reliable experimental data set, useful also to validate the results of computational fluid dynamics simulations, where ideal geometric boundaries are usually assumed.

For the first time, this work gives an insight into the behavior of the benchmark rectangular cylinder in realistic high-Reynolds-number turbulent wind conditions. Moreover, the effects of changing the angle of attack between 0° and 10° ; the Reynolds number in the range 11,800 to 112,600; and both the turbulence intensity and the integral length scale of the incoming flow are investigated. One of the goals of this work is to provide experimental results to fill in the gap evidenced by the synopsis by Bruno et al. (2014), in particular concerning surface pressure data at various small angles of attack, flow velocity measurements in the wake (as explicitly required by the benchmark (Bartoli et al., 2008) but at present still completely missing) and turbulent flow results in general. In addition, this paper provides a discussion of the mechanism of impinging shear-layer instability in smooth and turbulent flow.

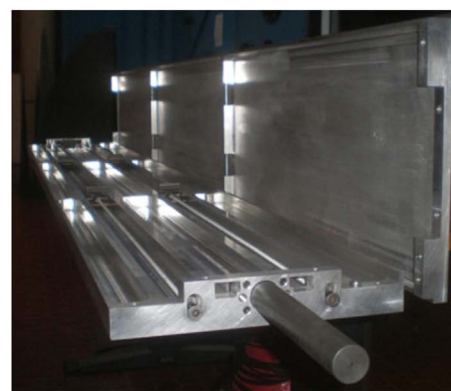
2. Wind tunnel tests

2.1. Experimental set-up

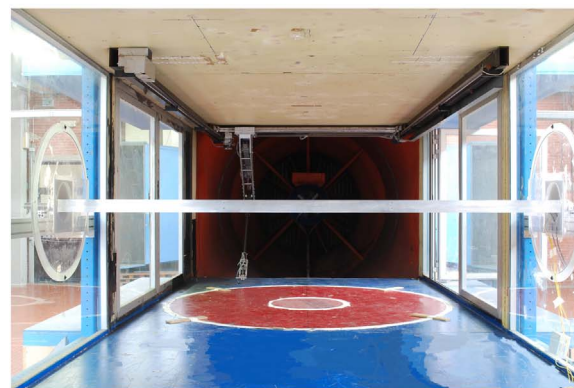
The tests were carried out in the open-circuit boundary-layer CRIACIV¹ wind tunnel in Prato, Italy. The facility is 22 m long and presents at the inlet a nozzle with a contraction ratio of 3 after the honeycomb and a T-diffuser at the outlet. The rectangular test section is 2.42 m wide and 1.60 m high. Air is drawn by a motor with a nominal power of 156 kW, and the flow speed can be varied continuously up to 30 m/s by adjusting through an inverter the rotation speed of the fan or the pitch of its ten blades. In the absence of turbulence generating devices, the free-stream turbulence intensity is about 0.7%.

The model was made of a special aluminum, for which the residual stresses were relieved to obtain a prismatic shape with very small tolerance. It was milled to obtain very sharp edges and smooth surfaces. The model was 300 mm wide (B), 60 mm deep (D) and 2380 mm long (L). Its length-to-width ratio L/B was 7.93 and its mass was 42 kg. The model locking system was activated from its ends, so to avoid screws and minimize the disturbances to the flow on the lateral surfaces of the prism (Fig. 1(a)).

The model was mounted horizontally in the wind tunnel, at mid-



(a)



(b)

Fig. 1. (a) View of the model and its internal structure; (b) model in the wind tunnel.

height of the test section, with its longitudinal axis perpendicular to the flow stream (Fig. 1(b)). It was rigidly connected to a steel frame placed outside the wind tunnel. At zero degree angle of attack ($\alpha = 0^\circ$), the short side of the cross section was perpendicular to the air stream and the solid blockage ratio was 3.75%. At the maximum angle of attack tested ($\alpha = \pm 10^\circ$), the blockage ratio was 6.95%. In any case, no blockage correction was applied to the data.

The ends of the model were very close to the wind-tunnel lateral walls but additional end-plates were installed to avoid pressure increase in the model near-wake regions. They were designed according to the principles reported in Cowdrey (1963), namely they were 480 mm $\times$ 300 mm (respectively in the streamwise and cross-flow directions), extending 30 mm in front of the leading edges and 150 mm behind the trailing edges of the model.

The model was instrumented with 564 pressure taps. However, during this test campaign only 61 taps in the midspan section were employed (Fig. 2). The pressure data were recorded with two 32-channels miniaturized piezoelectric scanners and the Pressure Systems' Digital Pressure Compensation (PSI DTC) Initium system, at a sampling rate of 500 Hz. The pressure signals were transferred to the scanners through 490 mm long Teflon tubes with an internal diameter of 0.8 mm, which were split at a distance of 110 mm from the scanner to insert a 20 mm long damper with an internal diameter of 0.3 mm. Thanks to this configuration of the tubes, no significant signal distortion was observed in the frequency range of interest.

The mean flow speed was monitored with a Prandtl tube placed 3.5 m upstream of the model and connected to a high-accuracy low-differential pressure transducer by Setra (AccuSense model ASL). Prior to model installation in the wind tunnel, the conversion coefficient between the mean dynamic pressure registered by the reference Prandtl tube and that measured at the midspan of the future model location was determined for each wind speed and free-stream turbu-

¹ Inter-University Research Center on Building Aerodynamics and Wind Engineering.

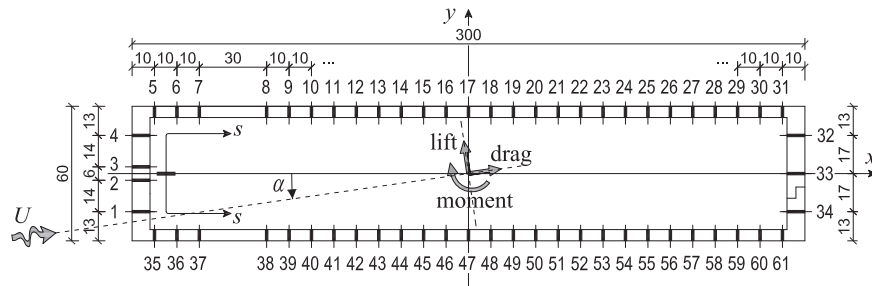


Fig. 2. Distribution of the pressure taps along the model cross section. The convention for reference coordinates, angle of attack and aerodynamic forces is also indicated. Dimensions are in mm.

lence considered. The same procedure was followed to determine the static pressure difference between the two locations. The flow angle of attack was adjusted by rotating the model about its longitudinal axis and by measuring the angle through a DXL360S inclinometer with an accuracy of 0.05°.

The characteristics of smooth and turbulent incoming flows and the fluctuating velocity field in the wake of the cylinder were determined through measurements at a sampling rate of 2000 Hz with a Dantec single-component hot-wire probe 55P11 connected to a Dantec CTA 56C01 module and a 56C17 bridge.

Finally, the measurement of the aerodynamic force coefficients at various angles of attack were performed by means of two high-frequency six-components strain-gauges force balances (ATI FT-Delta SI-165-15) placed one at each end of the model, externally to the wind-tunnel walls.

The high degree of symmetry of the model was verified by the results of preliminary tests reported in Bartoli et al. (2011). In that case, the model was rotated by 180° about its longitudinal axis and about the wind-tunnel axis, obtaining exactly the same pressure distribution for all of the four configurations. Furthermore, the lowest vertical bending and torsional mode frequencies of the system were respectively 20 Hz and 43 Hz. Consequently, flow speeds producing vortex shedding at about these frequencies were avoided to estimate the Strouhal number. Also, given that these natural frequencies are not high enough, the measurement through the force balances of the fluctuation amplitude of the force coefficients was avoided.

2.2. Turbulent flow characteristics

The nominally homogeneous isotropic turbulent flow was generated by two different bi-planar wooden grids variably distanced upstream of the model (Fig. 3), following the procedure of Laneville (1973). Grid A was characterized by a mesh of 100 mm×100 mm with 25 mm deep slats; grid B by a mesh of 550 mm×550 mm with slats of 140 mm (both grids had a thickness in the streamwise direction of 25 mm). The spectral properties of the free-stream flow at the position of the model were determined through hot-wire measurements. The spatial homogeneity of the flow at the position of the model was verified prior to starting the tests. The various configurations considered are listed in Table 1, where the turbulence intensity and the longitudinal integral length scale are reported. The latter was calculated according to Taylor's frozen-eddy hypothesis and the results were confirmed by fitting the expression proposed by von Kármán to the measured spectrum (see Fig. 4 for three examples of measured and fitted power spectral density).

3. Pressure measurement results

Pressure time histories were recorded for about 120 s, corresponding to about 11,400 and 56,100 nondimensional convective time units (U/D , where U is the incoming mean flow speed and t the physical time) respectively for the minimum and maximum flow speeds (and

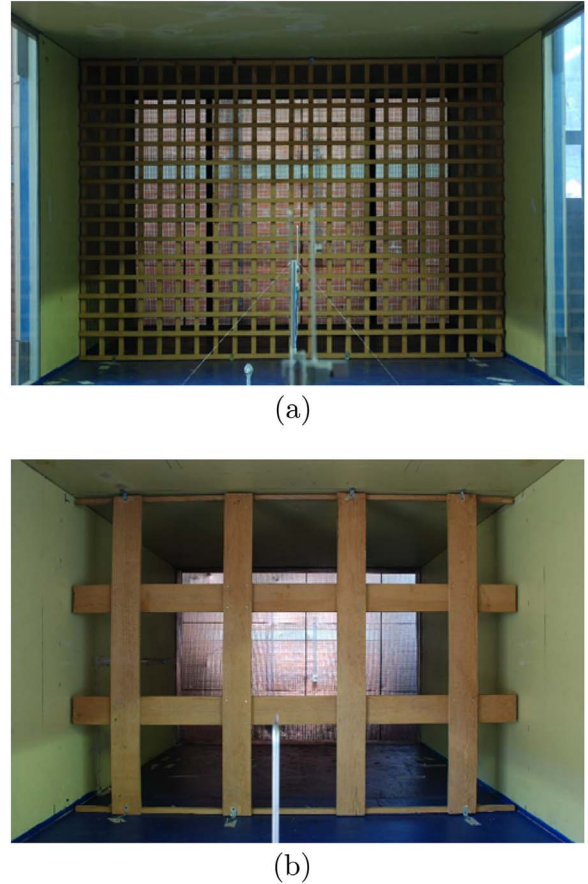


Fig. 3. Grids used to generate turbulence: (a) grid A; (b) grid B.

Table 1

Turbulent flow characteristics in the various tests performed. x_m denotes the distance between the grid and the model axis, M is the grid mesh size ($M = 100$ mm for grid A and $M = 550$ mm for grid B), I_u the turbulence intensity, and L_u^x/D the longitudinal turbulence integral length scale. α indicates the flow angles of attack tested.

Configuration	Grid	x_m	x_m/M	I_u	L_u^x/D	α
		[m]	[-]	[%]	[-]	[deg]
#1	No grid	—	—	0.7	—	−10 to +10
#2	A	4.95	49.5	2.9	1.3	0, +2
#3	A	1.50	15.0	5.7	0.7	0
#4	A	0.90	9.0	8.6	0.6	0
#5	B	7.14	13.0	8.8	4.4	0
#6	B	5.00	9.1	13.6	3.9	0

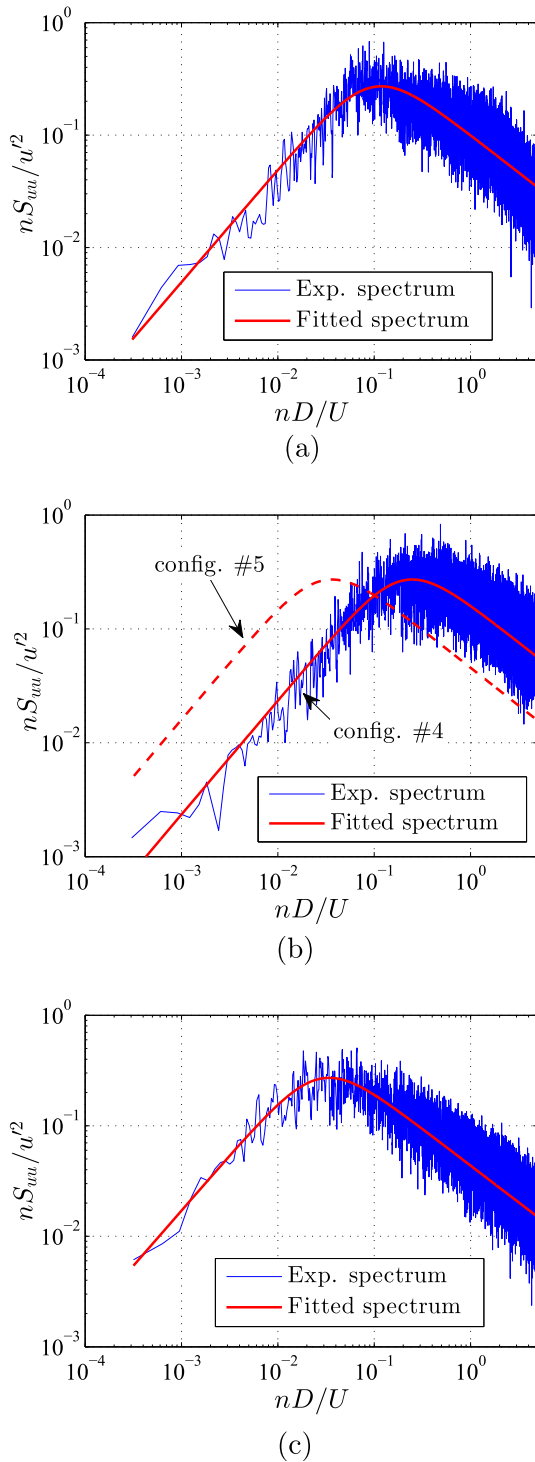


Fig. 4. Normalized power spectral density of longitudinal turbulence fluctuations measured with the hot-wire anemometer for three configurations tested (see Table 1): (a) configuration #2, $I_u=2.9\%$, $L_u^*/D=1.3$, $Re=48,700$ (based on model section depth D); (b) configuration #4, $I_u=8.6\%$, $L_u^*/D=0.6$, $Re=50,000$ (the comparison with the fitted von-Kármán spectrum for configuration #5 is also provided); (c) configuration #6, $I_u=13.6\%$, $L_u^*/D=3.9$, $Re=47,000$.

therefore Reynolds numbers) considered. The pressure signals were filtered to eliminate undesired noise. In particular, a 6th-order Butterworth low-pass filter with a cut-off frequency of 200 Hz was applied to all configurations to remove electrical disturbances at high frequencies. Moreover, a 6th-order band-stop filter was used to remove the narrow-band disturbance due to the rotation of the wind-tunnel motor fan. This filter had a bandwidth of 5 Hz, with limits depending

on the specific rotation frequency of the fan for each measurement. An additional 4th-order band-stop filter was applied between 4 Hz and 7.5 Hz to remove a spurious low-frequency peak in the pressure spectra, probably due to an acoustic resonance in the wind tunnel. The last filter was not applied in the test configurations with high turbulence (#5 and #6), since the energy content due to incoming turbulence was comparable or larger than that of the disturbance.

The pressure measurement chain was also affected by an intrinsic white noise, which can be considered as statistically independent from the physical fluctuating pressure. The no-wind measurements to quantify the white noise were carried out with the inverter of the motor fan switched on (but with negligible rotation speed and minimum blades' pitch), as the electrical noise was in this case slightly higher. The no-wind signals were treated with the previously mentioned filters, and then their energetic contribution was removed in terms of variance for each pressure tap. This operation is important especially for low-Reynolds-number measurements, i.e. at low flow speed, when the noise-to-signal ratio is higher. In particular, in this case, for the pressure taps on the windward face or those on the side faces close to the leading edges, the noise variance is comparable or even higher than the variance due to the pressure fluctuation. A similar denoising procedure was applied for the estimation of the integral forces from pressures, removing the contribution to the force variance due to the white noise.

Prior to starting the tests, the exact position of zero angle of attack ($\alpha=0^\circ$, see Fig. 2) was sought by comparing the pressure distributions on the upper and lower side of the instrumented section of the cylinder. A very good match was obtained by rotating nose-up the model to 0.45° .

3.1. Smooth flow

The Reynolds number $Re = UD/\nu$, where $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$ is the air kinematic viscosity, was varied between 22,800 and 112,200. Its effect on the mean pressure coefficient distribution was found to be negligible, as shown in Fig. 5. By contrast, a small difference can be appreciated in the root-mean-square (r.m.s.) peak value of pressure fluctuations (about -12% increasing Re from 56,700 to 112,200). The mean pressure in the first part of the separation bubble, close to the leading edge, takes values between 0.8 and 0.9. The pressure recovery starts around $s/D = 2.5$ (where s is the curvilinear abscissa defining the contour of the model cross section, as shown in Fig. 2), while the base pressure is -0.23 . The r.m.s. of pressure coefficient takes a value of about 0.04 in the region after the leading edges and reaches a maximum in the range 0.12 to 0.15 for s/D between 4 and 4.5.

The Strouhal number $St = n_s D/U$, where n_s is the frequency of vortex shedding, was determined from the frequency corresponding to the main peak in the spectrum of a pressure tap on the top face of the cylinder (tap 30, $s/D = 5.16$, see Fig. 2), in which the vortex-shedding contribution is clearly dominant. During the spectra calculation, the signals were windowed to obtain a resolution of 0.001 in terms of Strouhal number. As shown in Fig. 6, St resulted to be about 0.115 for a null angle of attack, which is coherent with the literature results collected in Mannini et al. (2011). A small increasing trend with the Reynolds number is suggested by the experimental results. The variation with the angle of attack is discussed in Section 3.3.

The inspection of the pressure power spectral density for all the pressure taps, summarized in Fig. 7 for the top side of the model, revealed that the great majority of the energy of pressure fluctuations is provided by the Strouhal peak. However, moving slightly downstream from the leading edge, the total variance remains approximately unchanged in a first portion of the lateral sides of the cylinder ($0.5 \leq s/D \leq 2$, see Fig. 5(b)) but the vortex-shedding peak progressively reduces, balancing the growth of the background component of the fluctuations. Prior to reaching the midchord, around $s/D = 2.5$, where the total variance has just started to increase, the spectrum is broad-banded and the Strouhal peak is hardly detectable. This means

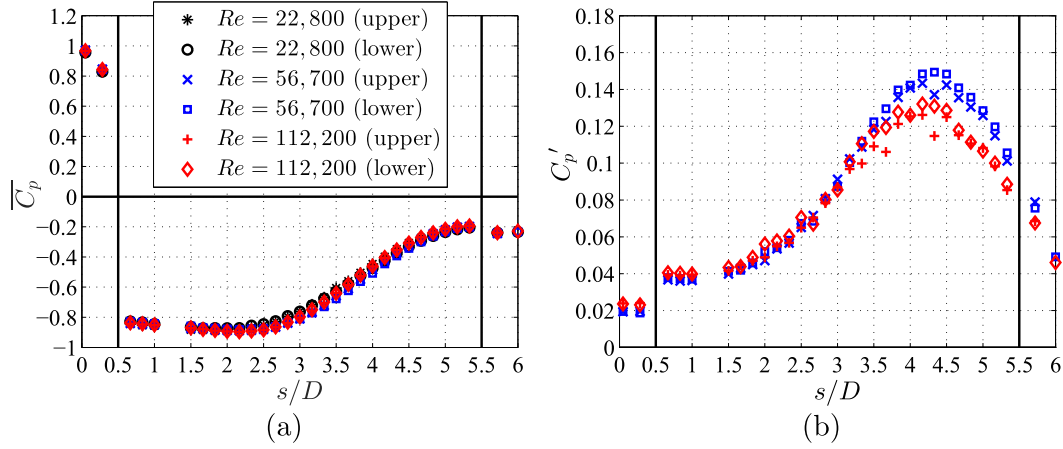


Fig. 5. (a) Mean and (b) r.m.s. values of pressure coefficients in smooth flow for $\alpha = 0^\circ$ (r.m.s. values for $Re = 22,800$ are not reported due to expected lower accuracy). $s/D = 0$ and $s/D = 6$ denote respectively the windward- and the leeward-face midpoints of the model.

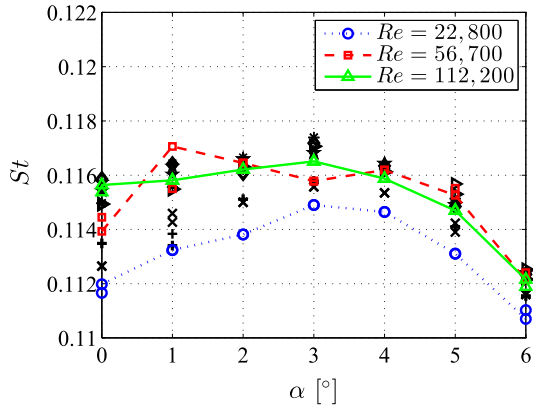


Fig. 6. Values of the Strouhal number for various angles of attack, detected from the spectra of the pressure coefficient of tap 30 ($s/D = 5.16$). All of the Reynolds numbers tested are reported but only three of them are indicated in the legend.

that after the leading edge, for a length of about $2D$, there is a progressive energy transfer from a narrow range to a wide range of frequencies. Going further downstream, the Strouhal peak becomes dominant again, particularly in the region of the maximum r.m.s. value. Then, approaching the trailing edge, it tends to decrease slightly. It is also worth noting that, close to the leading edge, the components at frequencies higher than n_s present very low energy.

Furthermore, the pressure time histories were integrated with the rectangle method to obtain drag, lift and moment coefficients (denoted respectively as C_D , C_L and C_M , all normalized with the section depth D). An example of lift coefficient time history is reported in Fig. 8(a), while the attendant probability density function (PDF) is shown in Fig. 8(b). It is clear that the PDF is not far from a Gaussian distribution, thus emphasizing the strong randomness of the narrow-band process.

3.2. Turbulent flow

Fig. 9 shows the mean and r.m.s. values of pressure coefficients for various turbulent flows. It is apparent that even a low free-stream

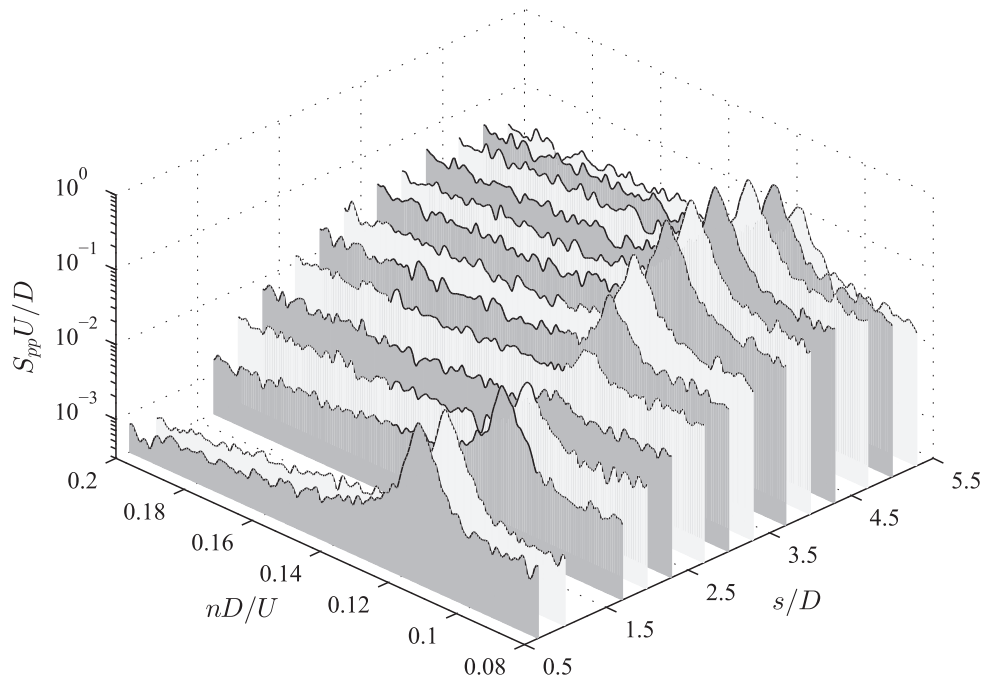


Fig. 7. Evolution of the dimensionless power spectral density of the pressure coefficient along the streamwise side of the cross section ($\alpha = 0^\circ$, $Re = 56,700$).

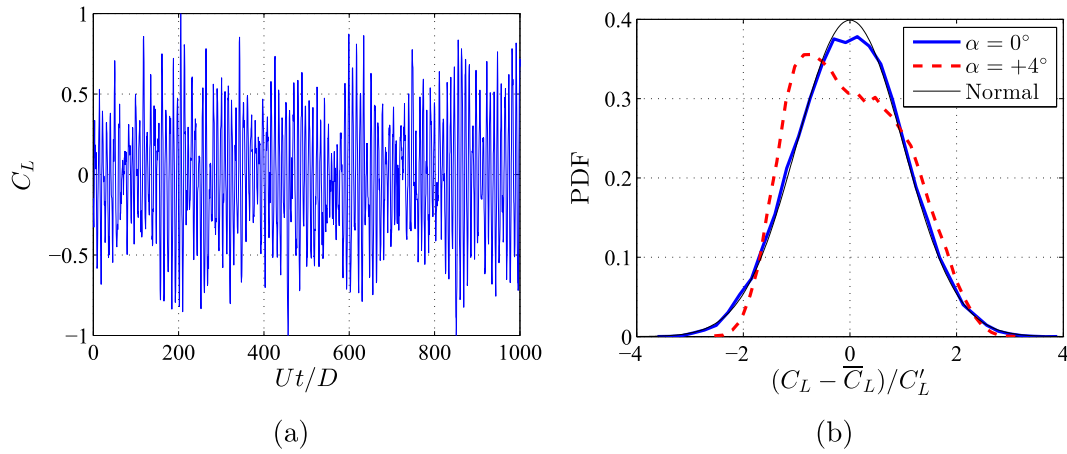


Fig. 8. (a) Time history of the lift coefficient for $\alpha = 0^\circ$ and $Re = 56,700$ (obtained through integration of pressures and normalized with the section depth D); (b) probability density function of the lift coefficient for $\alpha = 0^\circ$ and $+4^\circ$ ($Re = 56,700$ and $56,200$, respectively). $\bar{C}_L$ and C'_L denote respectively mean and r.m.s. values of the lift coefficient.

turbulence intensity (configuration #2) is able to promote an earlier reattachment of the shear layers and to reduce significantly the length of the separation bubbles on the lateral faces of the model, as shown by the faster recovery of mean pressures and the upstream shift of the r.m.s. peak, which is known to be related to the location of the mean reattachment point (Kiya and Sasaki, 1983). Higher turbulence intensities produce a more pronounced shortening of the separation bubble. In turbulent flow, the minimum pressure behind the front corners is also significantly reduced. This feature is fairly general in the case of blunt forebody with separating and reattaching flows, and Melbourne (1980) explained it with the stronger curvature of the shear layer due to free-stream turbulence and the related increase of the local velocity outside the separation bubble, which decreases the pressure on the surface. The effect of the integral length scale of turbulence L_u^x can be appreciated by comparing the results obtained for configurations #4 and #5, which both present a turbulence intensity $I_u \cong 8.7\%$ but values of L_u^x/D respectively of 0.6 and 4.4. Since the spectra of free-stream turbulence can be well fitted with a von Kármán model, an increase of L_u^x/D with constant I_u implies a reduction of the energy content corresponding to small-scale turbulence (see Fig. 4(b)). A higher turbulence length scale entails a slightly longer bubble length, a lower suction peak, and higher values of pressure fluctuations, in particular in the first half of the lateral sides of the cylinder, the latter effect probably due to the enhanced correlation of the unsteady incoming flow.

Fig. 10 reports the distribution of the third- and fourth-order

statistical central moments of pressures in the form of skewness (sk) and kurtosis (kt). Concerning the smooth flow case, the skewness is slightly positive immediately after the leading edges, becomes negative up to about $s/D = 3$, then positive and finally, close to the location of the mean reattachment, takes again negative values. Kiya and Sasaki (1983) showed that the range of positive skewness is related to the unsteady reattaching zone. They also conjectured that the positive spikes in the pressure time histories are associated with the inrush of outer irrotational fluid of high total pressure towards the surface under the action of large-scale vortices. High values of skewness and kurtosis are observed on the rear face of the cylinder, close to the trailing edges, even in smooth flow. In turbulent flows with small longitudinal integral length scale, the behavior is qualitatively similar to the smooth flow case, though the skewness is negative behind the leading edges and the points of sign inversion move farther upstream as the turbulence intensity increases. Quite different patterns, with large skewness and kurtosis, are observed for larger length scale of turbulence.

The integral mean drag force coefficient is shown in Fig. 11. Although the pressure taps on the windward and leeward sides of the section are too few for an accurate estimation of drag, it is clear that the effect of turbulence is small in this case, $\bar{C}_D$ passing from 1.13 to 1.21, with no effect of the change in the integral length scale. This result is in line with the data of Courchesne and Laneville (1982), which show a reducing sensitivity to turbulence of the drag force while the side ratio B/D is increased from 1 to 3. Nevertheless, a slight reduction of $\bar{C}_D$ with I_u was observed for the rectangular 3:1 section, while the opposite trend

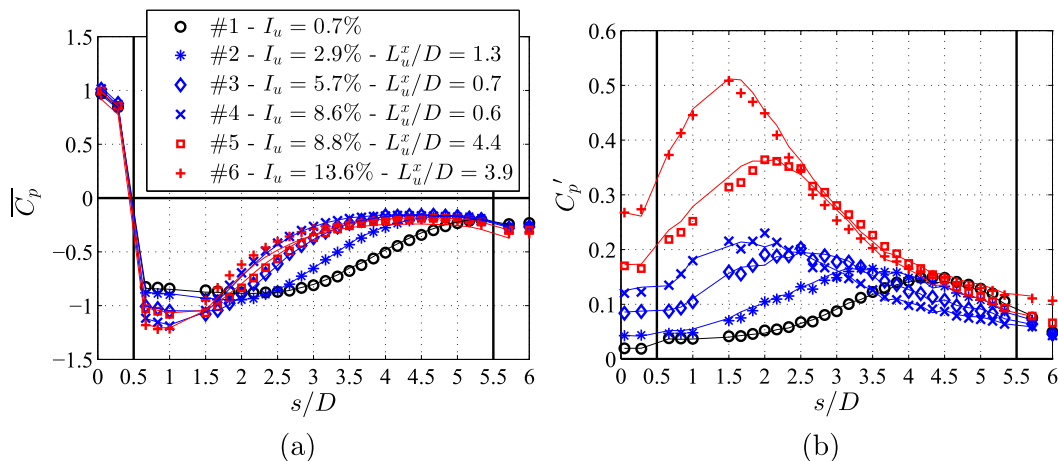


Fig. 9. (a) Mean and (b) r.m.s. values of pressure coefficients in smooth and various turbulent flows ($\alpha = 0^\circ$, $55, 900 \leq Re \leq 60,000$). Markers refer to the upper side of the section, while lines refer to the lower side.

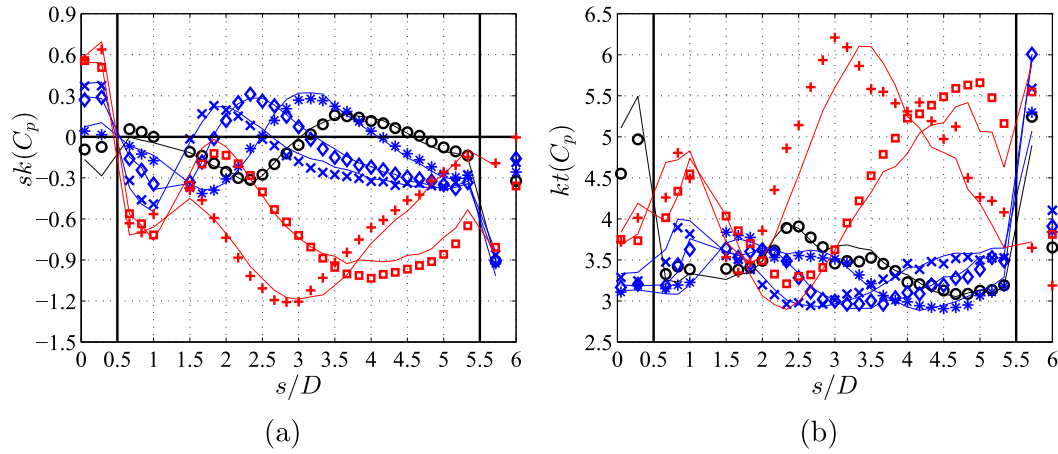


Fig. 10. (a) Skewness and (b) kurtosis values of pressure coefficients in smooth and various turbulent flows ($\alpha = 0^\circ$ and $55, 900 \leq Re \leq 60,000$). Markers refer to the upper side of the section, while lines refer to the lower side. Consider Fig. 9 for the legend.

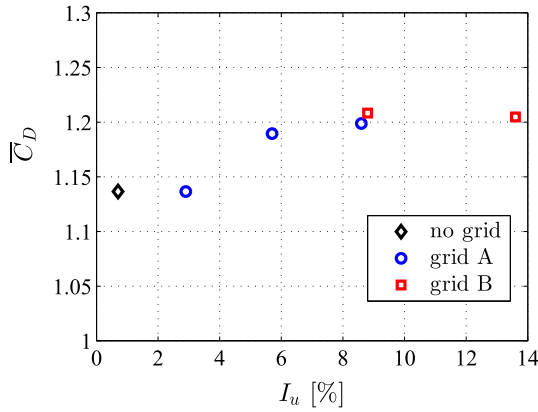


Fig. 11. Mean values of the drag coefficient (obtained through integration of pressures and normalized with the section depth D) at zero angle of attack in smooth and various turbulent flows ($55, 900 \leq Re \leq 60,000$).

is found here for the 5:1 section.

As shown in Fig. 12, in turbulent flow no sharp peaks appear in the pressure power spectral densities, and for some pressure taps the

energy contribution at the vortex-shedding frequency is overwhelmed by the background of the spectra. The Strouhal peak is more evident in the points of the long section sides near the trailing edge and in the base region. Also, for some turbulent flow configurations there is a slight difference between the spectral peak frequencies on the streamwise and leeward faces of the cylinder. Fig. 13 shows similar results for drag, lift and moment coefficients obtained through pressure integration, highlighting the broadness of the spectra in turbulent flow, especially in the case of larger integral scale. Fig. 14 summarizes the Strouhal number obtained in the diverse flow conditions from the frequency associated with the dominant peak in the spectra of the pressure records for two relevant taps and of the integral lift force (except in the case of configuration #6, for which a peak in the lift spectrum is not detectable). In spite of the previously mentioned broadness of the spectra in turbulent flow, a clear increasing trend of the Strouhal number with the free-stream turbulence intensity can be seen. In turbulent flow, the Strouhal peak is particularly evident in correspondence of the pressure taps in the last portion of the streamwise sides of the section (see e.g. Fig. 15), whilst a more complicated behavior is found in the cylinder forebody, where several competing peaks appear. In the moment coefficient spectra (Fig. 13), a small bump around a value of the reduced frequency of 0.2 can also be noted. This feature was observed for a few pressure taps as well, and it

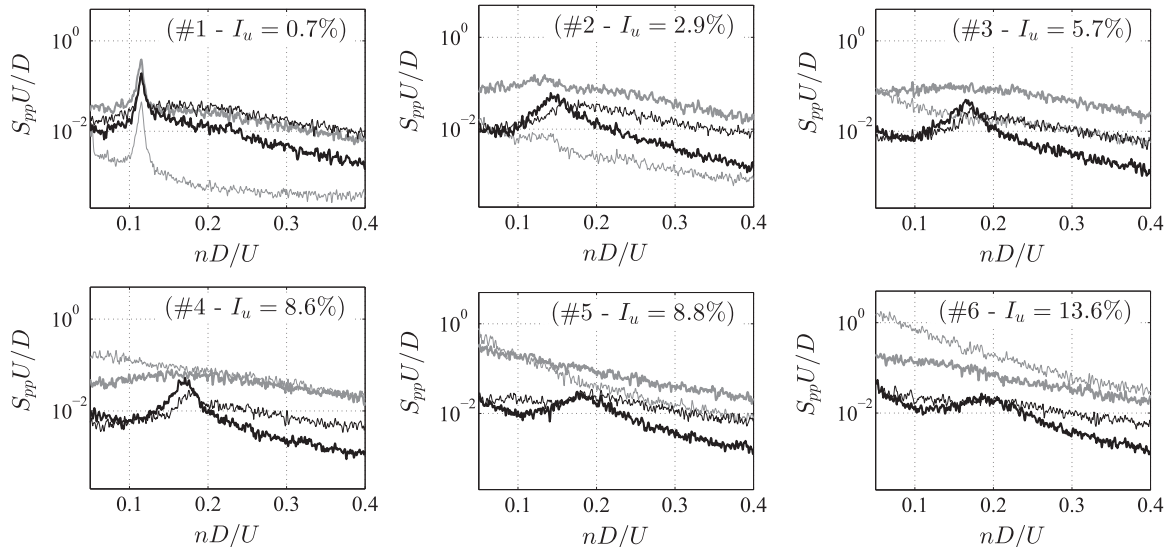


Fig. 12. Dimensionless power spectral density of pressure coefficient for tap 37 ($s/D = 1.0$, grey-thin), tap 49 ($s/D = 3.33$, grey-thick), tap 60 ($s/D = 5.33$, black-thin), and tap 34 ($s/D = 5.72$, black-thick), for various turbulent flows ($\alpha = 0^\circ$ and $55, 900 \leq Re \leq 60,000$).

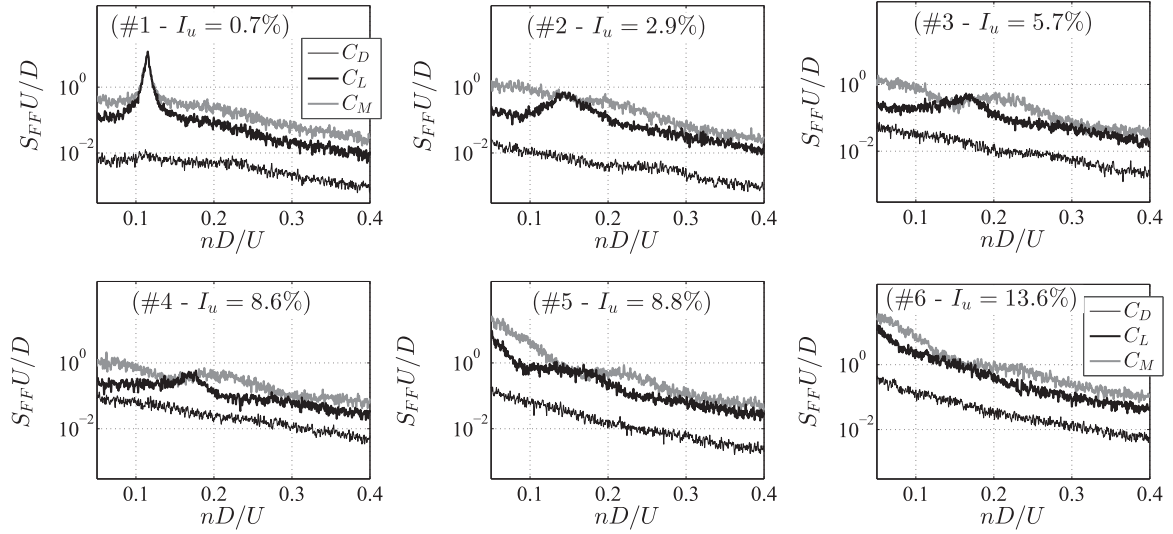


Fig. 13. Dimensionless power spectral density of force coefficients (obtained through integration of pressures) for various turbulent flows ($\alpha = 0^\circ$ and $55,900 \leq Re \leq 60,000$).

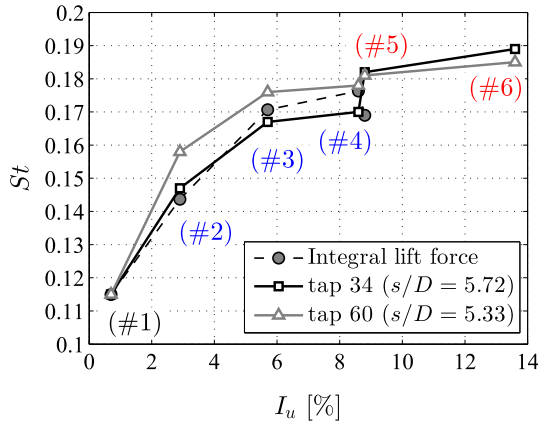


Fig. 14. Strouhal number estimated from the spectra of pressure coefficient at two tap locations and of integral lift coefficient for various turbulent flows ($55,900 \leq Re \leq 60,000$).

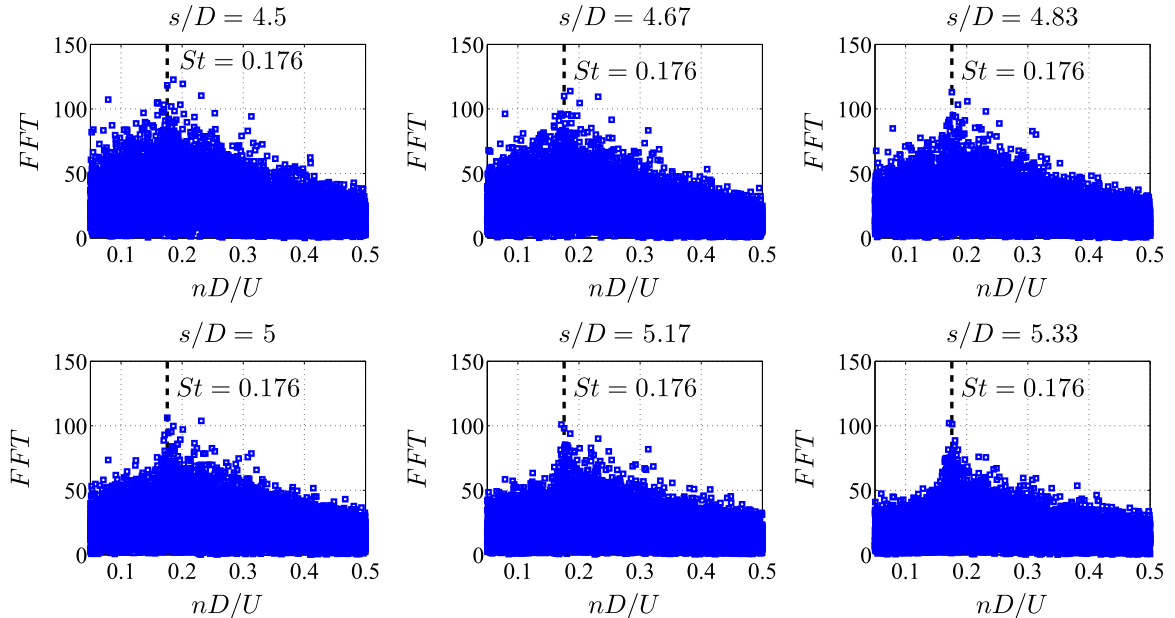


Fig. 15. Magnitude of the fast Fourier transform (FFT) of pressure records for some taps on the lateral side of the model (configuration #4, $I_u = 8.6\%$).

is probably related to a secondary mechanism of vortex shedding.

Fig. 16 depicts the phase angle of the pressure coefficient fluctuations at the dominant frequency of vortex shedding, relative to that of the tap closer to the leading edge (tap 5 in Fig. 2), along the lateral face of the cylinder. In smooth flow, the wavelength of pressure fluctuations is only slightly lower than the cross-section chord; that is, just one vortex travels at a time along the cylinder surface. This means that the first mode of impinging shear-layer instability is active in the vortex-shedding mechanism. This is in agreement with the numerical simulations reported in Mannini et al. (2010, 2011) and with the measurements of Nakamura et al. (1991), who interestingly showed that the vortices become two for $B/D = 6$. Fig. 16(a) also highlights that the phase angle is nearly null for $-2.5 \leq x/D \leq -1.5$, corresponding to the “pseudo-triangular region” described by Pullin and Perry (1980), and clearly evidenced by the numerical simulations of Bruno et al. (2010). Then, a progressive phase shift of 360° can be detected in the range $-1.5 \leq x/D \leq 2$, and finally a nearly constant phase is observed again for $2 \lesssim x/D \leq 2.5$, corresponding well to the “mean reattached flow region” identified by Bruno and co-workers. These considerations allow

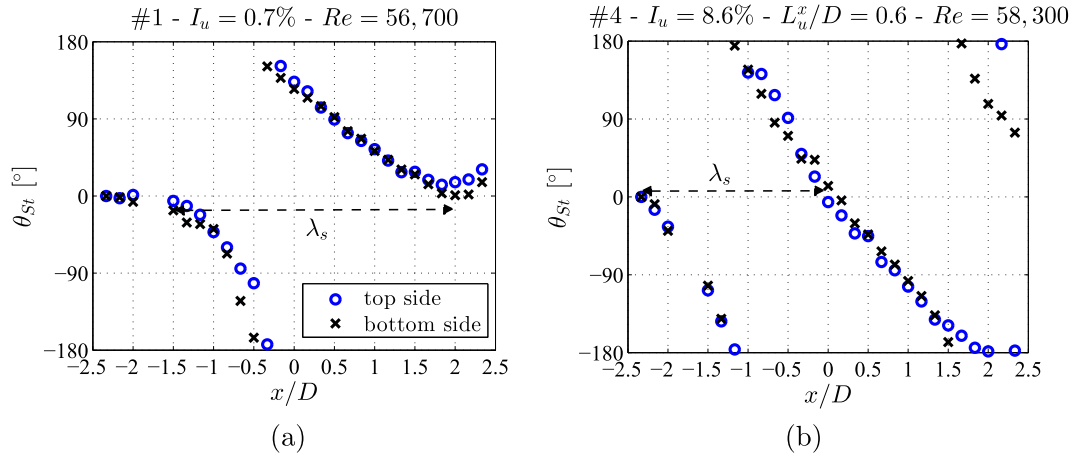


Fig. 16. Phase angle of the fluctuating pressure coefficient component corresponding to the Strouhal frequency at different streamwise positions: (a) smooth flow #1 ($St=0.115$); (b) turbulent flow #4 ($St=0.176$).

estimating the mean convection velocity U_c of the main vortices forming in the pseudo-triangular region, travelling downstream along the lateral sides of the cylinder, and finally shed in the wake. In fact, it is possible to write:

$$\frac{U_c}{U} = \frac{\lambda_s}{D} St \cong 0.4 \quad (1)$$

where U is the free-stream mean flow speed, $\lambda_s/D \cong 3.5$ is the normalized wavelength of the fluctuations at the vortex-shedding frequency n_s (Fig. 16(a)) and $St=0.115$ is the previously mentioned Strouhal number. The result of Eq. (1) indicates that the primary vortices travel along the lateral sides of the rectangular cylinder with a velocity significantly lower than the incoming flow speed. The gradient of the phase diagram in Fig. 16(a) suggests that the velocity of the primary vortices tends to increase in the range $-2.5 \leq x/D \leq -0.33$, while it is approximately constant for $-0.33 \leq x/D \leq 2$, where the phase-angle trend is linear. Similarly, Bruno et al. (2010) obtained $U_c/U = 0.48$ through LES simulation, while Nakamura et al. (1991) experimentally estimated $U_c/U = 0.63$ for a rectangular 4:1 cylinder.

By contrast, Fig. 16(b) clearly shows that in a turbulent flow with $I_u = 8.6\%$ and $L_u^x/D = 0.6$, the second mode of impinging shear-layer instability becomes dominant, as highlighted by the phase shift of 360° in about half a chord of the model section ($\lambda_s/D \cong 2.3$). This means that two vortices travel at a time along each lateral side of the cylinder, resulting in a significantly higher Strouhal number ($St=0.176$). The switch from the first to the second vortex-shedding mode due to free-stream turbulence is coherent with the previously discussed earlier reattachment of the separated shear layers, which promotes a vortex-shedding mechanism typical of a rectangular cylinder with a longer afterbody. Applying Eq. (1), one obtains again $U_c/U \cong 0.4$, indicating that the vortices are convected downstream at the same speed as in smooth flow, in spite of the significantly different mean flow features (Fig. 9). In a turbulent flow with higher intensity and larger integral length scale (configuration #6), the mechanism of vortex shedding is less clear. Diverse small peaks stand out from the background of pressure spectra in the various parts of the instrumented cross section, and it may be conjectured that the first and second modes of impinging shear-layer instability coexist due to the intermittency of the vortex-shedding process. Nevertheless, further investigations are required to ascertain it.

3.3. Effect of the angle of attack

Mean and r.m.s. values of pressure coefficients for various angles of attack in the range $0^\circ \leq \alpha \leq +6^\circ$ (nose-up rotations of the model, see Fig. 2) are shown in Fig. 17. A faster mean pressure recovery and an

upstream shift of the maximum r.m.s. of fluctuations can be observed on the bottom side of the cylinder (Figs. 17(c)-(d)), where an earlier time-averaged reattachment of the shear layer occurs due to the angle of attack. The opposite behavior is clear for the top side (Figs. 17(a)-(b)). The numerical simulations carried out by Mannini et al. (2010) and the comparison with the measurements of Schewe (2013) revealed that the mean flow is already fully separated from the top side for $\alpha = +4^\circ$. In addition, the comparison with the flow visualizations reported in Mannini et al. (2010) suggests that the sharp increase of fluctuating pressure immediately before the top trailing edge, observed for $\alpha = +4^\circ$ and even more for $\alpha = +6^\circ$, is an evidence of a small counter-rotating secondary bubble.

Although the separation bubble length and the surface pressures significantly change with the angle of attack, the mean pressure variation along the bottom face in correspondence of the separation bubble looks similar in the various cases. Such a behavior was observed by Roshko and Lau (1965) for diverse forebody shapes, and suggested to them to plot the pressure data in a normalized fashion. Robertson et al. (1978) applied this approach to the results for a square cylinder at several angles of attack. They scaled the coordinate describing the position of the pressure taps on the lateral face of the cylinder with the distance S_R from the leading edge of the mean reattachment point, and normalized the mean pressures (expressed as $1 - \bar{C}_p$) with the peak suction. In this way, an evident collapse of the pressure distributions was obtained in correspondence of the separation bubble, confirming that the flow has a similar structure in the bubble for different angles of attack, while the pressure distribution after reattachment follows a different trend as it is associated with a different phenomenon. Fig. 18 reports the universal plot obtained for the rectangular 5:1 cylinder analyzed in the present work. Due to the absence of detailed measurements for identifying the length of the bubble, the mean reattachment position was assumed to coincide approximately with the local maximum of the mean pressure coefficient along the lateral face of the cylinder. Indeed, Robertson et al. (1975, 1978) showed that mean flow reattachment occurs between the location of the peak of C'_p and the local mean pressure maximum, slightly before the position of the latter. Nevertheless, for the results at zero angle of attack a fictitious value of $5.5D$ was settled for S_R , since the bubble length is only slightly smaller than the streamwise dimension of the rectangular section and no local pressure maximum is found. The collapse of the mean pressure coefficients in the normalized plot of Fig. 18 clearly highlights the universal features of the separation bubble at various angles of attack for the rectangular 5:1 cylinder in a similar way to what previously observed for the square cylinder at incidences for which reattachment occurs. Although the results are not reported here in the interest of brevity, the application of the same procedure to the data for various

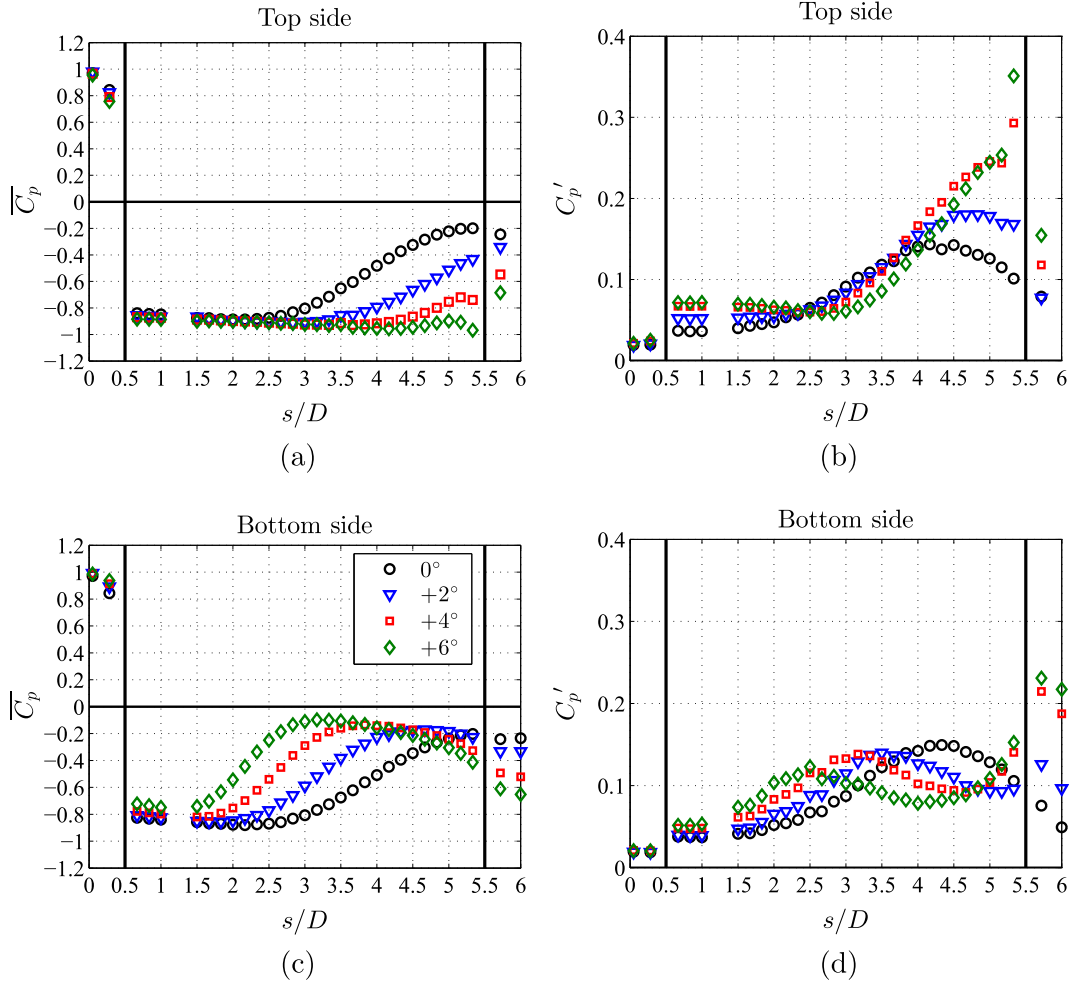


Fig. 17. Mean (left) and r.m.s. values (right) of pressure coefficients for various angles of attack in smooth flow ($56, 200 \leq Re \leq 56, 700$) for the top (a–b) and the bottom side (c–d) of the model.

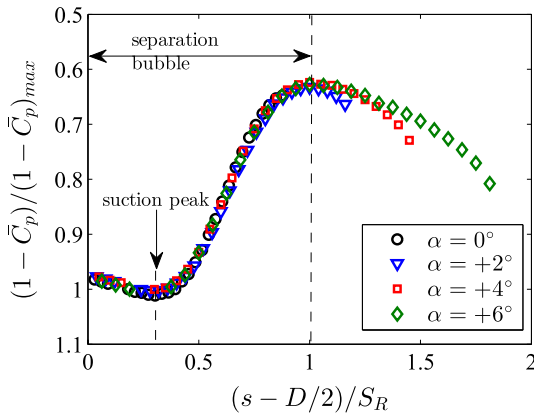


Fig. 18. Collapse of separation bubble pressure distribution for various angles of attack.

turbulent flows at zero angle of attack (Fig. 9(a)) showed that the self-similarity of the pressure distribution in correspondence of the separation bubble is maintained only for low turbulence intensity (configuration #2).

Fig. 19 reports the first two statistical moments of pressure coefficients for an angle of attack of $+4^\circ$, emphasizing Reynolds-number effects. In fact, the recovery of mean pressure is faster on the lower side in the range $1.5 \lesssim s/D \lesssim 3$ for $Re = 22,500$ as compared to $Re = 56,200$ and $111,200$, while the nearly constant pressure after

separation is slightly lower on the upper side. Differences can also be appreciated in the distribution of C'_p . These results are in line with the small Reynolds-number effects observed by Schewe (2013) in the range considered in the present tests. By contrast, stronger Reynolds-number effects are expected for $Re \lesssim 10, 000$ and $Re \gtrsim 200, 000$ (Schewe, 2013; Mannini et al., 2010). It is also to note that the self-similarity of pressures changing the angle of attack (Fig. 18) does not apply here for different Reynolds numbers, as the minimum and maximum pressures on the lower side of the section remain unchanged, whereas the recovery depends on the Reynolds number.

As evidenced by the pressure spectra in Fig. 20, a small angle of attack is able to foster sharper peaks at the Strouhal frequency, which do not disappear close to the midchord of the section as observed in smooth flow. In addition, in the base region both the peaks corresponding to the Strouhal number and, with less energy, to its first superharmonic are visible for $\alpha \geq 3^\circ$. In Fig. 6 it is apparent that the Strouhal number slightly tends to increase for α passing from 0° to 3° , and then to decrease in the range 3° to 6° . However, for $\alpha = 4^\circ$ it is larger than for $\alpha = 0^\circ$, while for $\alpha = 6^\circ$ it is smaller than for a null angle of attack.

Fig. 21 shows the drag, lift and moment coefficients obtained through integration of the pressures measured on the contour of the midspan section of the model. Reynolds-number effects on the lift and moment coefficients are evident for angles of attack larger than about 3° . This finding well compares with the results of Schewe (2013), who showed a slight reduction of the mean lift coefficient in the interval $20, 000 \lesssim Re \lesssim 100, 000$. By contrast, outside this range he observed a

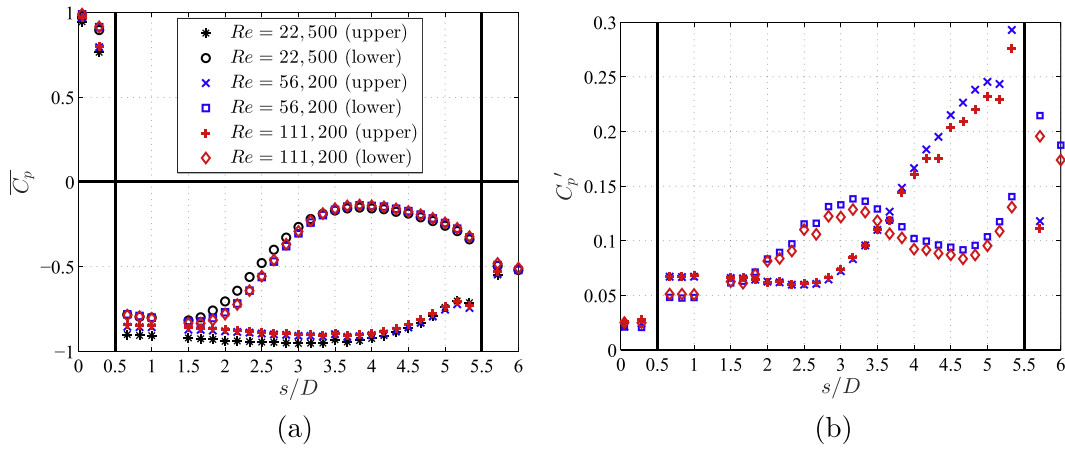


Fig. 19. (a) Mean and (b) r.m.s. values of pressure coefficients in smooth flow for $\alpha = +4^\circ$ (r.m.s. values for $Re = 22,500$ are not reported due to expected lower accuracy).

significant increase of $\bar{C}_L$. Larose and D'Auteuil (2006, 2008) observed similar Reynolds-number effects for rectangular cylinders with side ratios of 2, 3 and 4, and claimed that their behavior could be affected by Re because the airflow might not separate at the sharp edges with the same state, especially for angles of attack different from 0° . Table 2 reports the values of lift-coefficient slope in the origin (calculated in the range 0° to 1° , since the nonlinear pattern is clear already for $\alpha = 2^\circ$). A small constant increase of $dC_L/d\alpha(0)$ with the Reynolds number can be appreciated.

The right-hand side of Fig. 21 shows the significant increase of the r.m.s. value of all the integral force coefficients up to $\alpha = 5^\circ$. For a zero angle of attack, the measured values of C'_L are between 0.27 and 0.33. Schewe (2013) estimated a typical lift coefficient r.m.s. value of about 0.4. However, the present results refer to a single section and therefore do not account for the reductive effect of the limited spanwise correlation of force fluctuations (Matsumoto et al., 2003; Ricciardelli and Marra, 2008; Ricciardelli, 2010; Mannini et al., 2011; Bruno et al., 2012).

Finally, Fig. 21 shows that turbulence of low intensity does not change appreciably the mean drag for 0° and 2° , but significantly increases the lift and reduces the absolute value of the moment coefficient. This is due to the shortening of the separation bubbles on the lateral sides of the cylinder, so that a larger angle of attack is required in turbulent flow to produce the complete burst of one of

them. The r.m.s. values of forces seem to be similar to the smooth flow case, apart from a non-negligible reduction for the lift coefficient.

4. Force measurement results

The results of total force measurements carried out with two force balances in the range of angles of attack between -10° and $+10^\circ$ are shown in Fig. 22 in terms of mean values of drag, lift and moment coefficients (all normalized with the short side of the section D). Time-averages were done over records of 90 s. The drag coefficient at zero angle of attack is $\bar{C}_D \cong 1.05$, in agreement with other literature results reported in Mannini et al. (2011). By contrast, the values obtained by pressure integration (Fig. 21) are slightly larger than 1.1, likely due to the small number of pressure taps on the front and rear sides of the instrumented section. The lift and moment coefficients show non-negligible Reynolds-number effects for angles of attack larger than 3° , in agreement with pressure measurement data. Moreover, the results for angles of attack between 6° and 10° are very sensitive to small variations of flow and geometric boundary conditions, and indeed the lift curve is slightly non-symmetric in this range. As compared to the pressure integration results (Fig. 21), the agreement is good for the lift coefficient while it is only qualitative for the moment coefficient, once again due to the insufficient number of pressure taps in certain portions of the section contour.

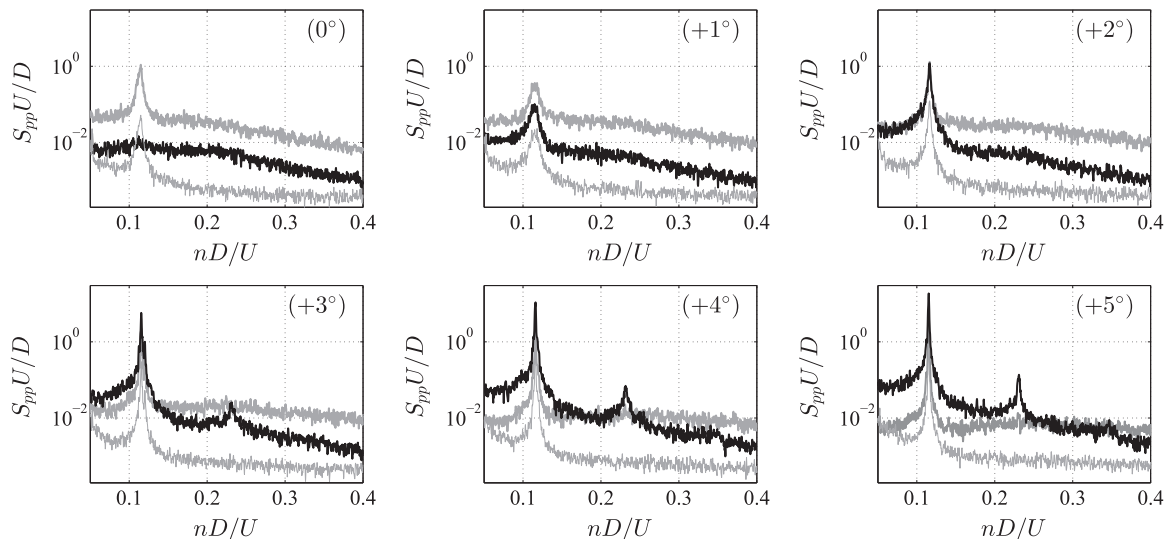


Fig. 20. Dimensionless power spectral density of pressure coefficient on the bottom side of the cylinder for various angles of attack in smooth flow ($56,200 \leq Re \leq 56,700$): tap 37 ($s/D = 1.0$, grey-thin), tap 53 ($s/D = 4.0$, grey-thick), and tap 33 ($s/D = 6.0$, black-thick).

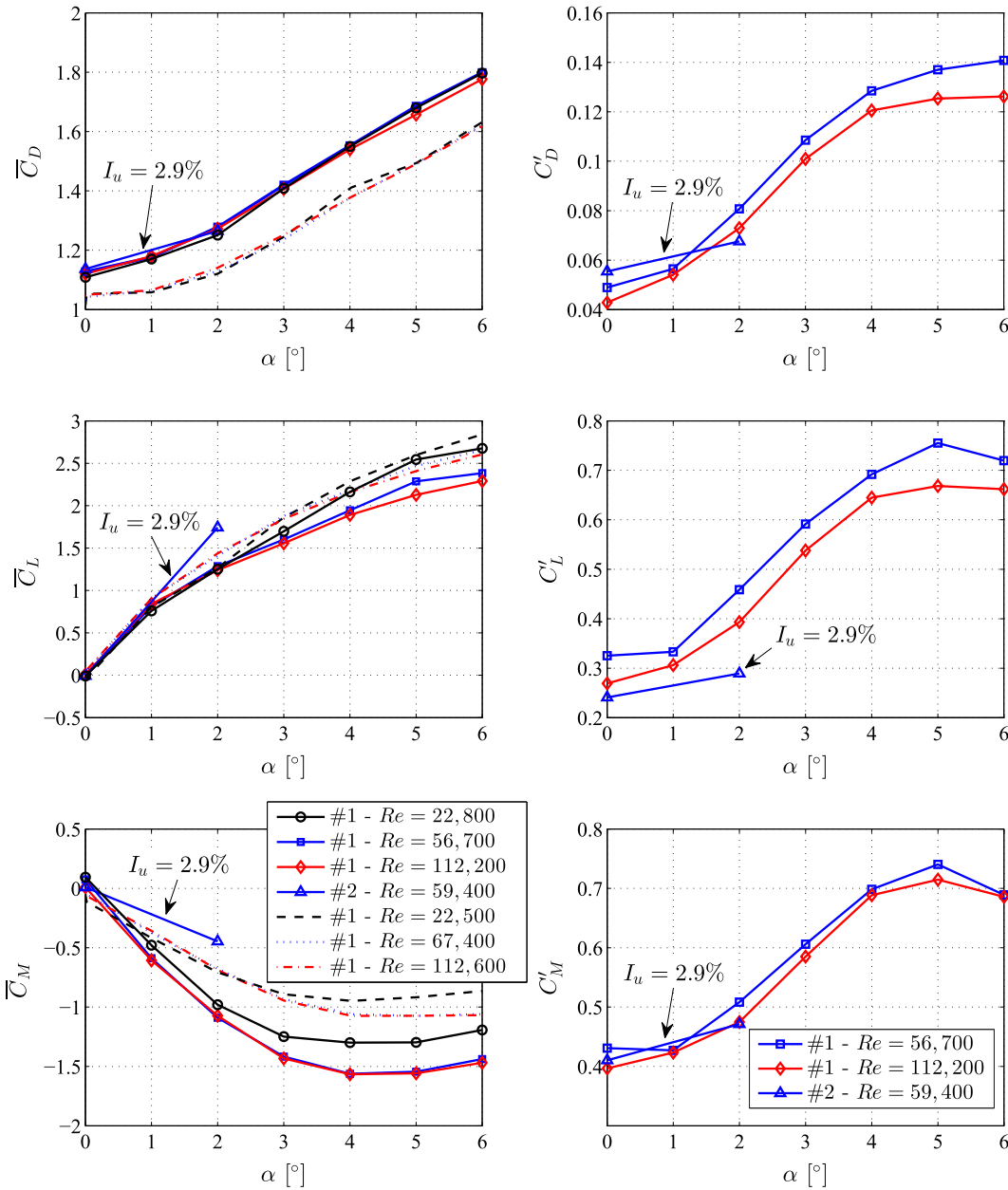


Fig. 21. Mean and r.m.s. values of drag, lift and moment coefficients (normalized with the section depth D) for various angles of attack. Lines with and without markers refer respectively to results obtained from integration of pressures (r.m.s. values for $Re = 22,800$ are not reported due to expected lower accuracy) and to direct measurements of forces.

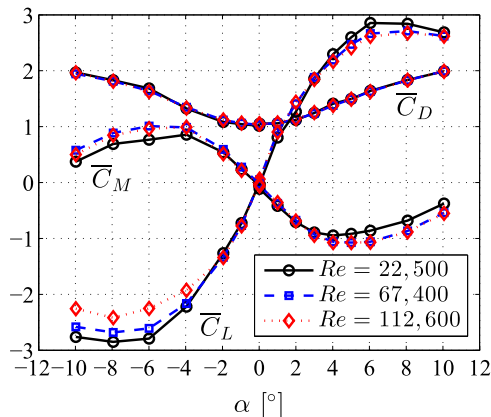


Fig. 22. Mean aerodynamic force coefficients determined through force measurements.

Table 2

Slope of the lift coefficient (obtained through integration of pressures and normalized with the section depth D) for $\alpha = 0^\circ$ and various Reynolds numbers.

Configuration	Re	$dC_L/d\alpha(0)$
	[-]	[-]
#1	22,800	43.5
#1	56,700	46.5
#1	112,200	48.3
#2	59,400	50.0

The value of about 0.115 of the Strouhal number for zero angle of attack is confirmed by the force measurements (Fig. 23(a)), as well as the significant decrease for flow incidences larger than 5° . Fig. 23(b) clarifies that the Strouhal number does not remain constant with the angle of attack if the dominant frequency in the lift spectra is normal-

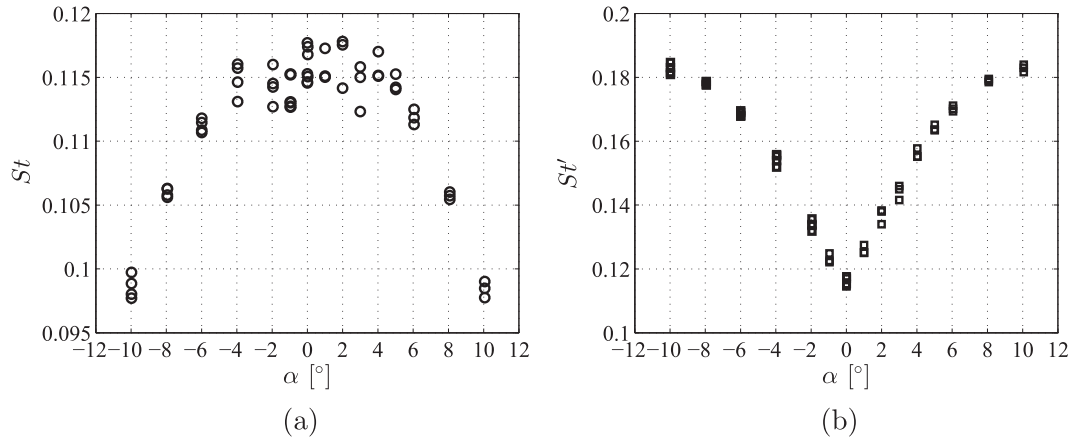


Fig. 23. Strouhal number determined from the power spectral density of measured lift force for various angles of attack and all of the Reynolds numbers considered: (a) normalization with the cross-section short-side length D ; (b) normalization with the cross-wind projection of the inclined section $D' = D \cos \alpha + B \sin \alpha$.

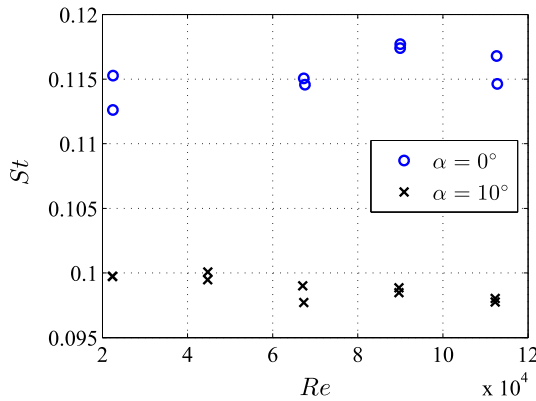


Fig. 24. Strouhal number vs. Reynolds number, obtained from force measurements for two angles of attack.

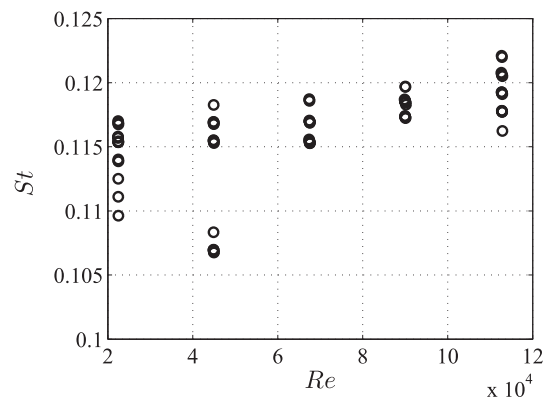


Fig. 26. Strouhal number values obtained from velocity fluctuations in the wake of the model (all along line $l6$ in Fig. 25).

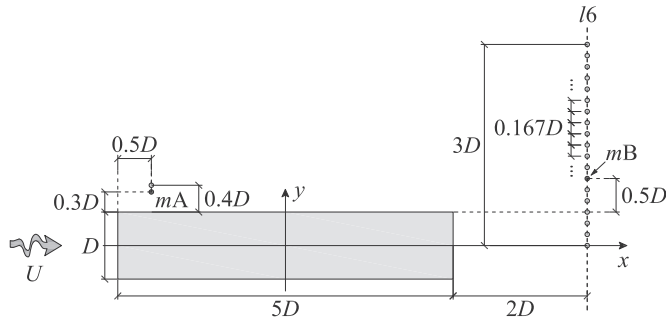


Fig. 25. Position of anemometry measurement points in the wake of the cylinder; $l6$, mA and mB represent respectively a reference transverse line and two reference points of the benchmark study (Bartoli et al., 2008).

ized with the cross-wind projection of the inclined section. In fact, in this way the re-defined Strouhal number is found to increase monotonically with the angle of attack. Finally, Fig. 24 shows the weakly increasing trend of the Strouhal number with the Reynolds number for $\alpha = 0^\circ$. The opposite light trend is observed for $\alpha = 10^\circ$.

5. Wake measurement results

Measurements with a hot-wire anemometer in the wake of the model were performed in smooth flow for $\alpha = 0^\circ$ around the positions denoted in Bartoli et al. (2008) as point mA (close to the free shear

layer separating from the leading edge) and along line $l6$ (in the wake of the cylinder, transverse to the incoming flow direction, at a distance $2D$ behind the rear face of the model). Fig. 25 indicates the position of the measurement points. Since the hot-wire probe was placed vertically, i.e. along the y -direction, the measured velocity has to be interpreted as the absolute velocity in the x - z plane. Velocity fluctuations were recorded for 90 or 120 s, corresponding to no less than 58,800 and 42,200 convective time units (Ut/D) respectively for $Re=11,800$ and 112,600.

The Strouhal number values obtained from velocity fluctuations along line $l6$ are reported in Fig. 26. In spite of the non-negligible dispersion of the results, in this case the tendency of the Strouhal number to increase with the Reynolds number is clearer than for pressure and force measurements.

Results for line $l6$ in terms of statistical moments of velocity fluctuations are shown in Fig. 27. No significant Reynolds-number effect was observed in the investigated range $22,500 \leq Re \leq 112,600$. Nevertheless, as previously mentioned, some Re -dependence of the wake characteristics may be expected for $Re \lesssim 10,000$ (Schewe, 2013; Mannini et al., 2010). The local mean flow velocity $\bar{u}$ is very low near the wake centerline ($0.33U$), and it quickly increases with the transverse distance in the range $0.33 < y/D < 1.5$ (Fig. 27(a)). For $y/D = 1.5$ the velocity recovery is nearly complete ($\bar{u}/U$ larger than 0.95). By contrast, the r.m.s. value u' of the velocity fluctuations reaches a maximum ($u'/U=0.195$) around $y/D = 0.67$ and then rapidly decreases (Fig. 27(b)). However, u'/U is still about 0.031 for $y/D = 2$ and 0.014 for $y/D = 3$. According to Bearman (1965), transversely with respect to the wake centerline, the point of maximum velocity fluctuation coin-

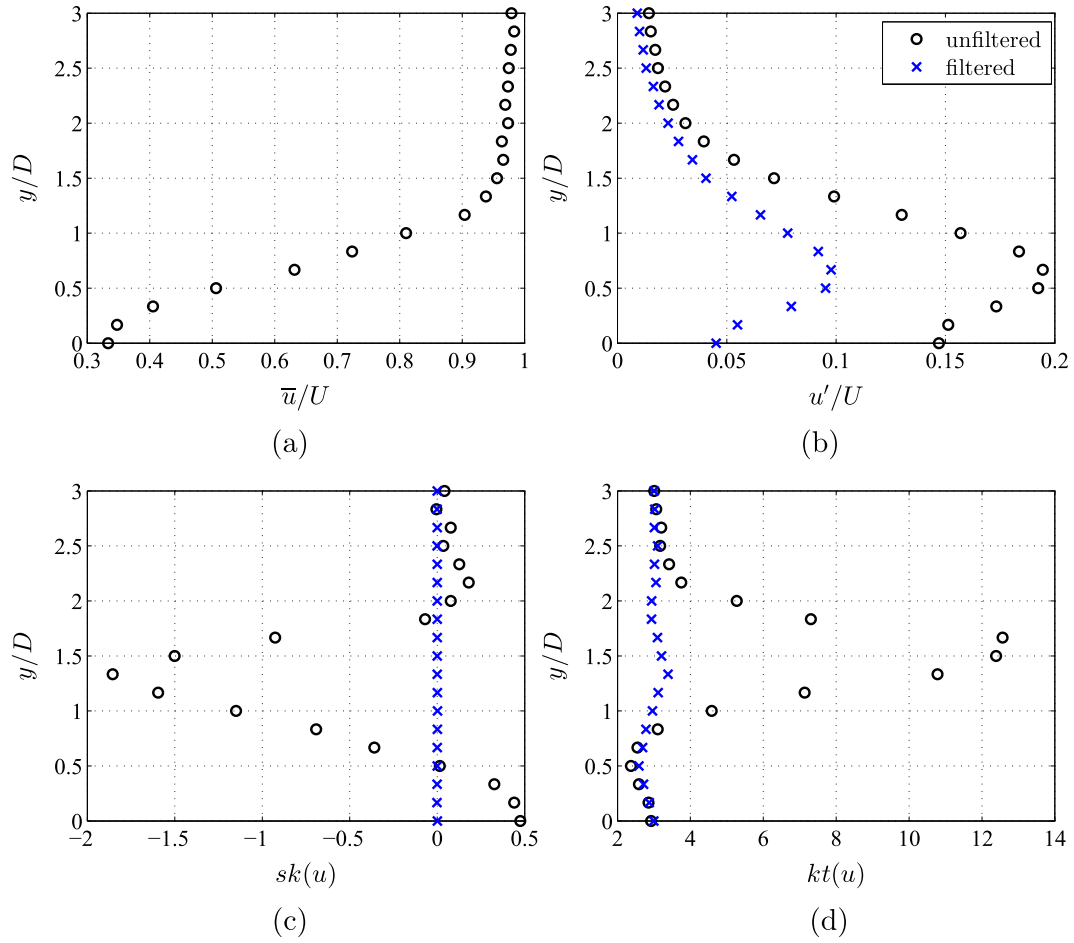


Fig. 27. Statistical moments of velocity fluctuations $u(t)$ along the transverse line l_6 , located at a distance equal to $2D$ downstream of the rear face of the model (Fig. 25). $y/D = 0$ denotes the wake centerline, while $y/D = 1$ corresponds to the point mB . The Reynolds number is 112,600. Results obtained after bandpass-filtering around the Strouhal frequency are also reported.

cides with the average location of the travelling vortex center. Concerning the first and the second statistical moments, a qualitative comparison can be carried out considering the results for the square cylinder reported in Bearman (1965), Durão et al. (1988) and Lyn et al. (1995).

The skewness of flow velocity fluctuations starts from a positive value of about 0.5 at the centerline, changes sign around $y/D = 0.5$ and reaches the minimum value of -1.85 for $y/D = 1.33$ (Fig. 27(c)). Then, it decreases down to zero but it also changes the sign again for $y/D \cong 2$. The kurtosis starts from a value of about 3 (similar to a normal

probability distribution) at the centerline, slightly decreases down to 2.4 for $y/D = 0.5$, then quickly grows reaching a maximum of about 12.6 around $y/D = 1.67$, and finally decreases to take again a value of about 3 for $y/D = 2.5$ (Fig. 27(d)). Similar patterns of third and fourth statistical central moments of velocity fluctuations were observed in the wake of a circular cylinder (Fabris, 1983; Petrović et al., 1998). In particular, Fabris (1983) explained the positive value of the skewness at the wake centerline as a consequence of eddies moving more vigorously toward the wake center than away from it. Fabris (1983) also clarified that the large values of skewness and kurtosis are due to

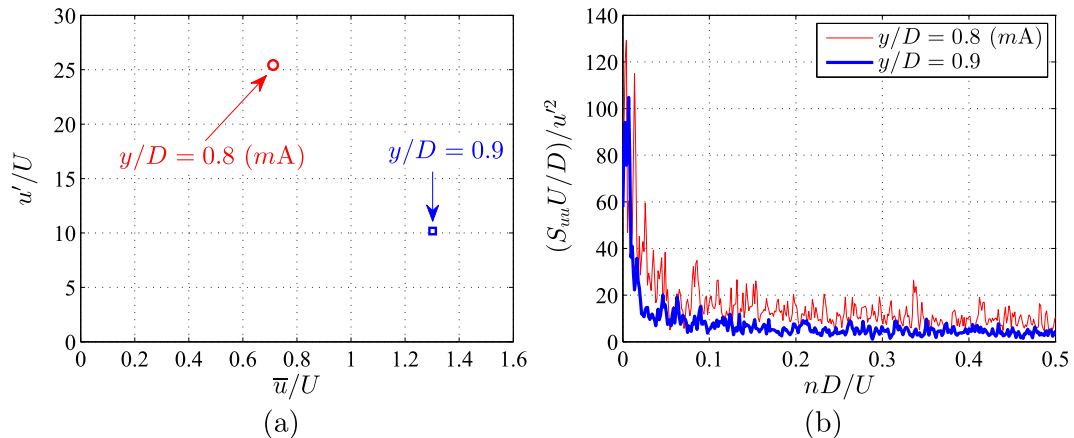


Fig. 28. (a) Statistical moments and (b) power spectral density of velocity fluctuations at the point mA ($y/D = 0.8$) and at a slightly larger distance from the model ($y/D = 0.9$, see Fig. 25). The Reynolds number is 11,800.

intermittency of the flow near the wake edge, where lumps of irrotational fluid are entrained in the turbulent wake (Pope, 2000). The location of skewness peak value has also been used for a conventional identification of the wake width (Mariotti and Buresti, 2013). Finally, the off-center kurtosis local minimum can be explained as an effect of recently entrained yet not-well-mixed irrotational fluid (Fabris, 1983).

The velocity fluctuation time histories were processed with a 3rd-order Butterworth bandpass filter around the Strouhal frequency ($0.08 < nD/U < 0.15$), and the results are shown together with the statistics of the unfiltered signals in Fig. 27. The variance of the fluctuations is significantly reduced at the centerline (u'/U below 0.05), where the Strouhal peak is even not present in the spectra. However, the pattern of u'/U is very similar to that previously described for the unfiltered records. In particular, the maximum value (u'/U around 0.1) occurs at the same location as the unfiltered data. Far away from the centerline, $y/D \geq 2$, the difference between filtered and unfiltered results is small (less than 0.006), meaning that the modest velocity fluctuations are almost exclusively at the Strouhal frequency. Along line l_6 , skewness and kurtosis of the filtered velocity fluctuations take always values close to 0 and 3, respectively.

The statistics of the hot-wire measurements at point mA ($x/D = -2$ and $y/D = 0.8$) and slightly above ($x/D = -2$ and $y/D = 0.9$) are reported in Fig. 28(a). Unlike the previously discussed tests, in this case only measurements at low flow speed, and therefore low Reynolds number ($Re = 11,800$), were possible due to the limited stiffness of the hot-wire supporting rig. Large gradients of the statistical moments are observed, indicating the presence there of the free shear layer (Bruno et al., 2010, 2014; Mannini et al., 2010, 2011). In particular, the local flow speed is lower than the free-stream velocity at $y/D = 0.8$ ($\bar{u}/U = 0.71$) and higher for $y/D = 0.9$ ($\bar{u}/U = 1.30$). Finally, Fig. 28(b) clarifies that the energy content of the fluctuations in both points is mainly concentrated at very low reduced frequency, while the Strouhal peak is not visible in the spectra. Low-frequency components were also found in the pressure signals of the fore taps of the instrumented cross section, although the Strouhal peak was apparent in that case. Clear identification of high-frequency fluctuations due to Kelvin–Helmholtz instability of the shear layer was not possible either; this may be due to the fact that the point mA is too close to the leading edge, still in the nearly-steady pseudo-triangular region (Bruno et al., 2010).

6. Closing remarks

The results of a wide experimental investigation on the aerodynamics of a rectangular cylinder with a 5:1 cross-sectional side ratio (BARC section) in smooth and turbulent flow were shown and discussed in this paper. The outcomes of pressure, force and wake velocity measurements were compared and cross-checked. The effect of the flow angle of attack was also carefully addressed. The work highlighted key features of the aerodynamic behavior of the cylinder that are not obvious at a first sight, such as the non-negligible Reynolds-number dependence of the force coefficients for $\alpha \gtrsim 3^\circ$ and of the Strouhal number even for a null angle of attack.

The influence of free-stream turbulence with various intensities and integral length scales was extensively studied. It was clearly shown that the length of the separation bubbles along the lateral sides of the cylinder significantly reduces due to the incoming turbulence. Moreover, the Strouhal peak in the force and pressure spectra becomes broader as compared to the smooth flow condition but does not disappear, even for high intensity of turbulence. This means that vortex-excitation is also possible in turbulent flow, all the more so that incipient resonant vibration can strengthen the vortex shedding in a similar way as acoustic resonance (Stokes and Welsh, 1986) or applied harmonic transverse velocity perturbation (Mills et al., 2002).

Furthermore, it was found that the primary-vortex convection mean velocity in turbulent flow is the same as in smooth flow. Nonetheless,

free-stream turbulence affects the vortex-shedding mechanism of impinging shear-layer instability by promoting the switch from a dominant mode to another. For instance, the first vortex-shedding mode is active in smooth flow, whereas the second mode can clearly be detected in one of the turbulent-flow configurations studied. It was clarified that the latter behavior, typical of a longer rectangular cross section, is coherent with the earlier shear-layer reattachment observed in turbulent flow. These results must be borne in mind while predicting possible flow-induced vibration of flexible cylinders.

Finally, the flow velocity measurements enabled the characterization of some parts of the wake of this bluff body, providing also useful data for validation of CFD calculations.

Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jweia.2016.12.001>.

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